General Questions

1. **What is this project about?**

   - This project focuses on analyzing stock price data and predicting future stock prices using machine learning models. It involves data preprocessing, exploratory data analysis (EDA), feature engineering, and model training.

2. **What datasets did you use?**

   - I used stock price historical data, which typically contains features like Open, High, Low, Close, Volume, and Date. The dataset can be sourced from financial APIs like Yahoo Finance or Kaggle.

3. **What was the objective of your project?**

   - The main objective was to predict the future closing prices of stocks based on historical data and evaluate the performance of different predictive models.

Technical Questions

4. **What preprocessing steps did you perform?**

   - I handled missing values, normalized the data, and converted date-time features. Additionally, I created lag-based features to capture temporal dependencies.

5. **Which models did you use for prediction?**

   - I used machine learning models such as Linear Regression, Decision Trees, Random Forest, and LSTM (Long Short-Term Memory) for time-series forecasting.

6. **Why did you choose LSTM for this project?**

   - LSTMs are well-suited for time-series data as they can capture long-term dependencies and patterns, which are crucial for predicting stock prices.

7. **How did you evaluate the performance of your models?**

- I used metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) to evaluate the accuracy of predictions.

Algorithm and Implementation Questions

8. **What challenges did you face in this project?**

- Challenges included handling noisy and non-stationary stock price data, feature selection, and ensuring the model generalized well to unseen data.

9. **What libraries or tools did you use?**

- I used Python libraries such as NumPy, pandas, Matplotlib, scikit-learn, TensorFlow/Keras, and Jupyter Notebook for implementing the project.

10. **How did you split the dataset?**

- The dataset was split into training, validation, and test sets, typically in ratios like 70:15:15. I also used time-based splitting to avoid data leakage.

Domain-Specific Questions

11. **How did you address the non-stationarity of the data?**

- I applied techniques like differencing and log transformation to make the data stationary, as required by certain models.

12. **What feature engineering techniques did you use?**

- I engineered features such as moving averages, RSI (Relative Strength Index), and lagged values of prices to enrich the dataset.

13. **How did you ensure your model is not overfitting?**

- I used techniques like cross-validation, regularization, and early stopping during training.

Advanced Questions

14. **Can you explain the intuition behind LSTM cells?**

 - LSTMs have a memory cell that retains information over time and gates (input, forget, and output gates) that control the flow of information, making them effective for sequential data.

15. **What are the limitations of your approach?**

 - The model relies heavily on historical data and cannot account for sudden market shocks or news that might influence stock prices unpredictably.

16. **How would you improve this project?**

 - I would integrate sentiment analysis of news headlines and social media data to enhance predictions. Additionally, I could explore ensemble models for better accuracy.

Behavioral Questions

17. **What did you learn from this project?**

 - I gained practical experience in handling time-series data, implementing machine learning models, and understanding the complexities of financial data.

18. **What was the most exciting part of this project?**

 - The most exciting part was seeing the model predict future stock prices with reasonable accuracy and interpreting the patterns it captured.

19. **How do you think this project is relevant to your role as a data science intern?**

 - This project demonstrates my ability to work with real-world datasets, apply machine learning techniques, and draw actionable insights, which are crucial for a data science role.